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Lesson 8 Automated Accessibility Testing

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Scoping Information

Organization Tested: Reddit

Homepage URL: https://www.reddit.com/

Browser: Google Chrome

Operating System: Windows 11

Assistive Technology: NVDA

Automated Testing Tools: Lighthouse, Axe Dev Tools, Siteimprove Accessibility Checker

Important Note: Reddit is a social media website that updates its content continuously, Therefore, the code is not static and the findings may vary over time.

Automated Tools Scan Reports

Siteimprove Accessibility Checker

Total Issues (as listed): 11

Total reported findings across 11 categories: 109

Siteimprove Report	
Finding Category	**Total Amount of Findings**
Link missing a text alternative	11
Interactive element does not meet enhanced size	2
Inline frame missing a text alternative	1

Siteimprove Report	
Visible label and accessible name do not match	1
ARIA attribute unsupported or prohibited	48
Invalid state or property	6
Headings are not structured	1
Text not included in an ARIA landmark	25
Container element is empty	1
Line height is below minimum value	12
Presentational element is exposed to assistive technologies	1

Axe Dev Tools

Total Issues: 23

Axe Dev Tools Report	
Findings by Severity Level	**Total Amount of Findings**

Axe Dev Tools Report	
Critical	11
Serious	12
Moderate	0
Minor	0

Finding Category	**Total Amount of Findings**
Elements must only use supported ARIA attributes	9
ARIA attributes must conform to valid values	2
Frames must have an accessible name	1
Links must have discernible text	11

Lighthouse

Accessibility Score: 80

Lighthouse Report	
Finding Category	**Total Amount of Findings**
ARIA	2
Contrast	1
Names & Labels	1
Best Practices	2
Audio & Video	1

Finding Category	**Total Amount of Findings**
Elements must only use supported ARIA attributes	9
ARIA attributes must conform to valid values	2
Frames must have an accessible name	1
Links must have discernible text	11

Other Results	**Total Amount of Findings**
Additional Items to Manually Check	10

Passed Audits	24
Not Applicable	32

Compare and Contrast Lighthouse, Axe Dev Tools, and Siteimprove

When the Site Improve accessibility checker reports its findings, it organizes them into different categories as well as which success criteria failed and which conformance level it falls under. For remediation, it provides a short explanation as to why it is an issue and includes evidence by linking to its help center with a related article about the issue as well as links for information on the success criteria straight from w3.org. Also, it provides general tips on how to resolve these types of issues through its design kit feature.

Similarly, Axe Dev Tools organizes its reported findings into different categories. However, they categorize them by their level of severity. When you expand a particular finding, it sometimes states which standard it violates, such as WCAG 4.1.2, EN 301 549, or Section 508. It also records the exact time the issue was discovered by you. Furthermore, it gives you the option to visually highlight the finding on the page so you

can see exactly where it is. For remediation, it includes a snippet of the code for some of the findings along with a description of what needs to be fixed. Similar to Site Improve, it includes an external link related to the finding. However, the links are from Deque University.

Similar to Axe Dev Tools and Site Improve, Lighthouse organizes its reported findings into different categories. However, it groups them under clear and short labels such as "Names and labels," "Best practices," and "Audio and video." Axe Dev Tools and Site Improve use sentences to group and label the findings. Another similarity to Axe Dev Tools is that the findings are highlighted when expanded except that they are images that have a highlighted border where the finding is on the page. For remediation, similar to Site Improve, it includes an external link related to the finding. However, the links are from Deque University just like Axe Dev Tools.

Overall, the automated tool that detected the most issues is Site Improve as it found 109 failures. However, it displayed a total count of 11 issues, which can be confusing because it counts categories rather than individual issues. For example, "Link missing a text alternative" is one category, but it accounts for 11 issues. Thus, the total number of

reported issues across all categories was 109. The tool that reported the least issues is Lighthouse, which detected 7 findings. Most of the issues reported by these tools were related to ARIA attributes and links. Some issues I identified during manual testing, such as focus order and keyboard accessibility problems, were not reported by the automated tools. Yet, I did find some of these issues when conducting a manual audit. I found Siteimprove easier to use for a beginner because it connects findings to WCAG success criteria and provides tips on how to resolve issues. Axe Dev Tools is a more advanced tool as programming knowledge is required to understand what went wrong with the code. Lighthouse might be confusing as well for beginners since it includes code.

Manually Test the Findings

Site Improve

Flagged as a finding: *2.5.3 Label In Name: Visible Label and accessible name do not match. "Apple logo sign in with Apple vs Apple sign in with Apple". Using two different names for a single page element can create a confusing user experience for assistive technology users.*

Manual Test Results: Siteimprove flagged the "Sign in with Apple" button because its visible label and accessible name supposedly do not match. However, upon manual inspection with NVDA, the screen reader announces, "Clickable link, Sign in with Apple, button." The Apple logo isn't announced because it is decorative, not an interactive element. Additionally, the button's text clearly communicates its purpose. Therefore, this is a false positive and doesn't need to be fixed.

Axe Dev Tools

***Flagged as a finding**: Links must have discernible text: Ensure Links have discernible text.*

```
<a class="no-underline hover:n..." href="https://alb.reddit.c..." target="_blank" rel="noopener nofollow sp..." role="link" data-ad-click-locati...="credit_bar_whitespac..." data-ad-click-action="1" tabindex="-1">
```

Manual Test Results: Axe Dev Tools flagged all ad links because the links do not provide sufficient information. Upon manual inspection with NVDA, the screen reader announces the links as "Ad Link". Since the user is unable to tell what kind of ad is in the link, this finding is considered a true positive as it fails 2.4.4 Link Purpose (In Context).

Lighthouse

Passed Audit: *Image elements have [alt] attributes. Informative elements should aim for short, descriptive alternative text. Decorative elements can be ignored with an empty alt attribute.*

Manual Test Results: Lighthouse passed "Image elements have [alt] attributes." However, upon manual inspection with NVDA, there are copious violations of 1.1.1 Non-Text Content. In my Reddit Audit report, I provided evidence how NVDA uses the text on the post as alternative text for images in the post. This tool was clearly programmed to detect if alt attributes were included but not programmed to check if the alt text provided is meaningful and descriptive for screen readers. Therefore, this is a false negative issue that must be fixed.

Conclusion

Personally, I prefer Axe DevTools because it allows me to group findings by severity. This helps me prioritize critical issues first, followed by issues with lower severity. I also like that I can view a snippet of the code associated with a finding instead of manually searching through the developer tools and reading through the entire page's code. Another feature I like is that I can highlight a finding to visually see where it is located on the page, which makes investigating the issue easier.

As for the Siteimprove Accessibility Checker, I like that it correlates its findings with WCAG conformance levels and success criteria. I also appreciate that it allows me to filter the search results by conformance level. However, I think the way it counts the issues is confusing because the "total issues" counts the categories of findings rather than individual findings. For example, I initially thought Siteimprove had detected 11 issues when there were actually 109.

As for Lighthouse, I like that it includes a section showing which automated audits passed. I use these results as a guide for manual testing because passing an automated audit does not necessarily mean that an element is fully accessible. This helps me identify areas that may require further investigation. I also like that Lighthouse includes images of the findings with a highlighted border showing where the issue is located on the page.

All in all, automated testing should not be the only method used to evaluate accessibility. It is a useful starting point that helps identify potential issues and can be used to guide you to areas that might need more investigation. However, manual testing

is still necessary because automated tools have limitations and may miss accessibility barriers. Automated testing is limited because it can only test what they have been explicitly programmed to look for. For example, how I mentioned that Lighthouse passed alt text that wasn't accessible. This is because the tool was not programmed to check for meaningful alt text descriptions. Combining automated testing with manual testing with keyboard and assistive technology testing provides a more complete evaluation of a website's accessibility.